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Capture window and robustness of nonlinear-kicker injection into a 4 GeV storage ring with a small dynamic aperture

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Abstract: A single nonlinear kicker magnet (NKM) can inject off-axis without disturbing the stored beam, but its kick varies across the injected bunch, and fourth-generation storage rings offer little dynamic aperture to absorb the resulting spread. We quantify this trade-off for a wire-based NKM in a 4 GeV ring whose on-momentum horizontal dynamic aperture at the injection point extends from −12 to +13 mm, using the RADIA kick map, a transfer line tracked particle by particle from the booster, a 2 m septum-to-kicker drift and native nonlinear ring tracking. A linear one-turn-map model overestimates capture by tens of percentage points. Capture is highest when the incoming angle is cancelled close to the kick-map extremum, where the kick gradient across the bunch is smallest: at the selected operating point 91.1 % of the injected particles survive 500 turns (87.0 % with a thick-magnet model), compared with 94.6 % for a uniform kick of equal strength, and the angular window is about ±0.5 mrad. For an aligned kicker the stored-beam centroid moves by less than 10^{-5} of its rms size; because the field is cubic near the axis, a 0.2 mm offset between kicker and stored orbit raises it to 0.10 of the rms size. Under a combined error budget the mean capture drops to 37.8 % (95 % interval 33.1–42.7 %) with the thin kick and to 47.4 % (42.4–52.4 %) with the thick magnet, which is less sensitive to injection-angle errors. Capture is limited by the injected-orbit error at the septum that booster extraction jitter produces: correcting the static trajectory raises it by only 3.0 percentage points, whereas removing the jitter raises it by 13.0 points.

Keywords: Beam dynamics; Accelerator modelling and simulations (multi-particle dynamics, single-particle dynamics); Instrumentation for particle accelerators and storage rings - high energy (linear accelerators, synchrotrons)

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1 Introduction

Top-up injection into low-emittance storage rings requires that the stored beam is not disturbed while charge is added. A conventional four-kicker orbit bump leaves residual oscillations of the stored beam because the kicker pulses cannot be matched perfectly. A single pulsed multipole magnet whose field vanishes on the stored-beam axis avoids this problem: pulsed sextupole injection was demonstrated at the KEK Photon Factory [1], a non-linear kicker (NLK) with zero central field and an off-axis field maximum was developed for BESSY II [2, 3], and a multipole injection kicker (MIK) installed at MAX IV and SOLEIL reduces the stored-beam perturbation below 2 % of its rms size [4].

The price of transparency is a kick that varies across the injected bunch. Liu *et al.* [5] identified the nonlinearity of the kick at the injected-beam position, together with the injected-beam size, as the main cause of reduced injection efficiency. The effect is aggravated in fourth-generation lattices, whose strong sextupoles restrict the dynamic aperture (DA) available for the kicked bunch, and beam-based studies at BESSY II reached up to 97 % efficiency only with optimised sextupole settings [3].

This article quantifies that trade-off for a specific case: a wire-based NKM computed with RADIA [6], placed 2 m downstream of the injection septum of a 4 GeV storage ring whose on-momentum horizontal DA at the injection point is only about 12 mm. We ask how wide the injection window is, how much capture is lost relative to a uniform kick, how transparent the NKM is for the

stored beam, and how robust the window is against realistic errors. The injected beam is tracked particle by particle from the booster through a matched transfer line (BTS), and capture is evaluated with native nonlinear ring tracking. The study is simulation-only; it does not claim experimental or operational validation.

2 System and model

2.1 Storage ring and dynamic aperture

The ring lattice (circumference 799.30 m, 3483 elements) is tracked with the Accelerator Toolbox [7, 8] in six dimensions with RF and radiation pass methods. At the kicker centre, which is the injection, kick and observation point, $\beta_x = 16.27$ m, $\alpha_x = -0.145$, $\beta_y = 5.60$ m and $D_x = 0.3$ mm. The tunes are 68.18/23.26, the chromaticities +6.0/+3.4, the equilibrium emittance 61.6 pm rad and the rms energy spread 1.26×10^{-3}. The synchrotron period is about 290 turns and the horizontal damping time about 3950 turns. The lattice contains no physical apertures; losses are produced by the septum blade and by dynamical instability.

Figure 1 shows the DA obtained by native tracking of single particles over 500 turns. On momentum, particles with $x' = 0$ at the kicker centre are stable between −12 and +13 mm (1 mm grid); the stable range is ±14 mm at $\delta = -1$ % and shrinks to ±3 mm at $\delta = -3$ %. The corresponding horizontal acceptance is about 9 μm rad.

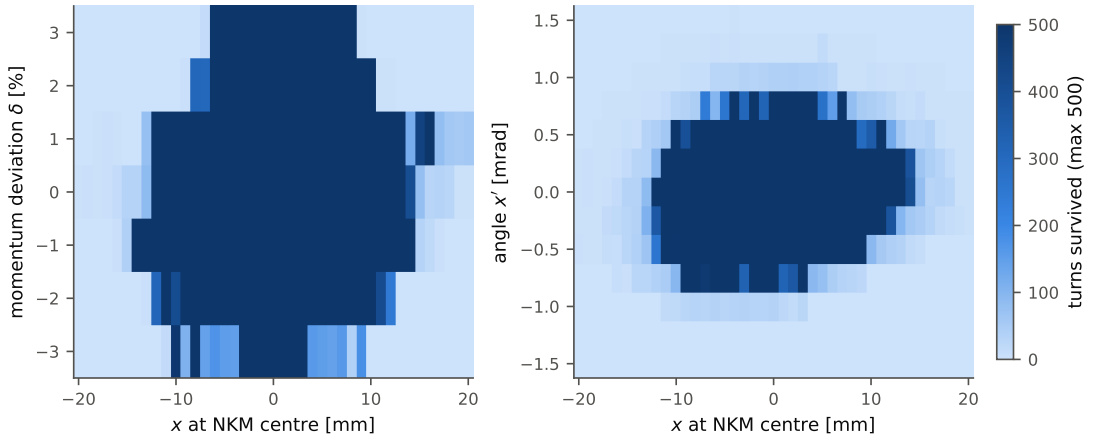


Figure 1. Dynamic aperture at the NKM centre from native element-by-element tracking of single particles for 500 turns (1 mm grid in x): turns survived versus horizontal position and momentum deviation at $x' = 0$ (left) and versus horizontal position and angle at $\delta = 0$ (right).

2.2 Nonlinear kicker field map

The NKM is formed by four current filaments of length 525 mm. RADIA field integrals along the beam axis, $\int B\,\mathrm{d}l$, divided by the magnetic rigidity $B\rho = 13.343$ T m at 4 GeV, give the kick map on a 201×201 grid over $|x|, |y| \leq 50$ mm (0.5 mm spacing). On the midplane the integrated kick equals the central field profile $B_y(x)$ multiplied by the magnet length to within 10^{-10} relative, the field vanishes on axis (kick $< 10^{-12}$ rad) and grows as x^3 near the axis. The horizontal kick reaches

its extremum of 5.75 mrad at $x = -8.5$ mm (figure 2). The kick map carries no charge sign; the polarity is a choice of coil current. We use the polarity for which the kick at $x < 0$ is negative, so that it cancels the positive angle of a beam travelling from the septum towards the axis. A field scale S multiplies the map. Coordinates outside the map are counted as losses; the map is never extrapolated.

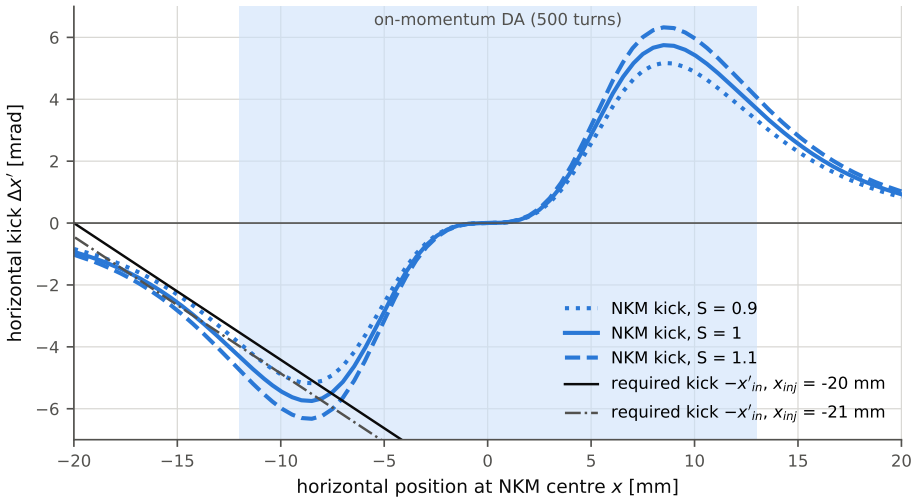


Figure 2. Horizontal NKM kick at $y = 0$ for field scales $S = 0.9$, 1.0 and 1.1 (blue), compared with the kick required to cancel the incoming angle of a beam launched from the septum at $x_{\mathrm{inj}} = -20$ and -21 mm with angle $x'_{\mathrm{in}} = (x - x_{\mathrm{inj}})/L$, $L = 2.26$ m (black). Intersections are angle-matched operating points. The shaded band is the on-momentum dynamic aperture.

2.3 Injection geometry and transfer-line handoff

The injected bunch is generated at the booster extraction point with $\beta_{x,y} = 7.56/12.27$ m, geometric emittances 100/10 nm rad and rms energy spread 1.1×10^{-3}, and is tracked element by element through the 21.79 m BTS line with the matched quadrupole settings (exit mismatch $\mathcal{M}_{x,y} = 0$, peak $\beta_{x,y} = 31.4/50.7$ m) or the baseline settings. At the septum exit the BTS distribution is shifted to the ring frame with centroid $(x_{\mathrm{inj}}, x'_{\mathrm{inj}})$. The septum blade has its stored-beam edge at -16 mm and a thickness of 2 mm; it removes injected particles on the wrong side at creation and circulating particles with $x < -16$ mm on every turn. The bunch drifts 2.26 m to the NKM centre, receives one thin kick (single-turn pulse) and is tracked for 500 turns. Capture is the fraction of particles alive after 500 turns, with 95 % Wilson intervals [9]. For each (x_{inj}, S) the matched angle cancels the kick at the nominal centroid; the angle offset from this value is scanned separately.

Two controls receive the identical injected particles. The *dipole* control is a uniform kick equal to the NKM kick at the nominal centroid, and the *linearised* control is the first-order Taylor expansion of the map at that point. The stored-beam response is evaluated from the exact kicks applied to a 10^5-particle equilibrium beam: the centroid betatron amplitude $A = \beta_x |\langle \Delta x' \rangle|$ is reported relative to the rms beam size $\sigma_x = 31.7\,\mu$m.

3 Model validation

Native element-by-element tracking is the reference. A linear one-turn map with the same septum check reproduces small-amplitude orbits (relative residual below 2 % at 1 mm after 10 turns) but has no DA. On identical injected bunches it predicts 62 % capture where native tracking gives 43 % (figure 3, provisional point $x_{\mathrm{inj}} = -20\,\mathrm{mm}$, $x'_{\mathrm{inj}} = 5.75\,\mathrm{mrad}$, $S = 1$), and for the dipole control the difference exceeds 40 percentage points. The linear map is therefore used only as a cross-check. Capture converges in the number of turns: 0.44 after 500 and 0.43 after 2000 turns. Increasing the particle number from 200 to 1000 changes the estimate within its interval. With zero field scale the NKM reproduces the kicker-off case exactly. A uniform 1 μrad kick on the stored beam gives a tracked centroid amplitude of 15.7 μm versus $\beta_x \Delta x' = 16.3\,\mu\mathrm{m}$, which validates the stored-beam estimate. A chamber aperture of $\pm 12 \times \pm 6$ mm at every element lowers capture by about 3 percentage points.

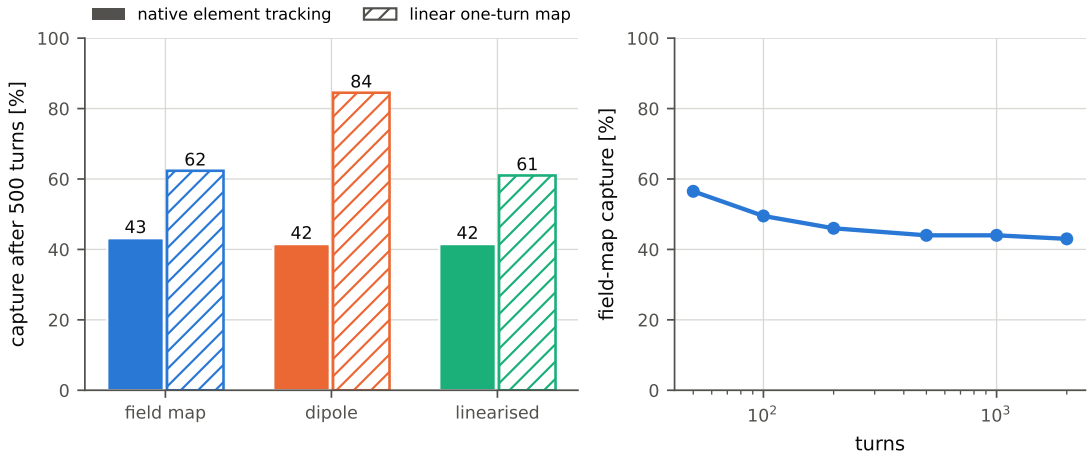


Figure 3. Left: capture after 500 turns at the provisional point for native element tracking (filled) and the linear one-turn map (hatched) on identical 200-particle bunches (field map: mean of three bunches). Right: field-map capture versus the number of turns (native tracking).

4 Injection operating window

Figure 4 maps the field-map capture over x_{inj}, S and the angle offset (200 particles per point, coupled matched-BTS bunch, native tracking). With the kicker off no particle is captured at any point, because the incoming angle of 4.9–6.2 mrad carries the bunch outside the DA. Capture is highest when the matched centroid lies close to the kick-map extremum: the selected operating point $x_{\mathrm{inj}} = -20\,\mathrm{mm}$, $S = 0.85$, $x'_{\mathrm{inj}} = 4.87\,\mathrm{mrad}$ (maximum lower confidence bound) captures 89.5 %. Increasing S moves the matched centroid towards the axis, where the kick gradient across the 0.7 mm wide bunch is larger, and capture decreases monotonically. The angular window is narrow and asymmetric: an offset of +0.5 mrad reduces capture to 13–64 %, −0.5 mrad to 15–85 %.

Figure 5 compares the models on four independent 500-particle bunches at the operating point. The NKM captures 91.1 % on average, the linearised control 96.2 % and the uniform dipole 94.6 %;

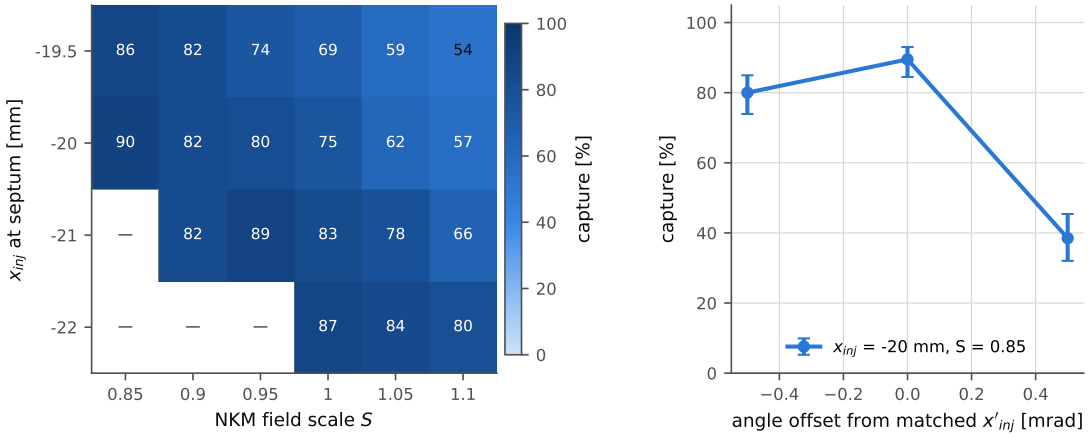


Figure 4. Left: field-map capture (%) at the angle-matched condition versus septum position x_{inj} and field scale S; dashes mark combinations without an inner matched solution inside the DA. Right: capture versus the angle offset at the selected operating point with 95 % Wilson intervals. Native tracking, 200 particles, 500 turns.

the kicker-off case captures 0.0 %. The capture deficit of the NKM relative to the uniform kick is the cost of the kick variation across the bunch. In return, the aligned NKM displaces the stored-beam centroid by 5×10^{-6} rms sizes and increases its emittance by a fraction 2×10^{-5}, whereas the dipole and linearised controls displace it by 2503 and 2971 rms sizes. Because the field grows as x^3 near the axis, the disturbance depends on the offset between the kicker centre and the stored orbit: 0.10 rms sizes at 0.2 mm and 1.5 at 0.5 mm. The stored-beam response uses a bicubic spline through the kick-map nodes; bilinear interpolation between the nodes at 0 and 0.5 mm would overstate the near-axis kick. With the baseline BTS optics the same NKM setting captures only 26.9 %, so matching the transfer line is a prerequisite. The thin-kick approximation of the 0.525 m magnet was checked by integrating the same bunches through the field profile in 160 drift–kick–drift slices: the thick magnet gives 87.4 % at the same injection angle and 87.0 % when the angle is re-matched (4.64 mrad), because the trajectory inside the magnet samples a steeper part of the kick profile. For the nominal beam the thin-kick capture is therefore optimistic by about 4 percentage points; section 5.2 shows that the opposite holds under errors.

5 Robustness

5.1 Transfer-line optics

With the matched BTS optics fixed (figure 6), three independent ensembles of 50 000 error realizations and one of 400 000 (quadrupole gradient 0.1 %, offsets 100 μm, roll 0.5 mrad, energy 0.1 %, booster β mismatch 5 %) give a median horizontal exit mismatch of $\mathcal{M}_x = 2.086\times10^{-3}$ (pooled 95 % interval 2.074–2.099$\times10^{-3}$). Only 5 of the 550 000 realizations exceeds the 60 m β limit, corresponding to a failure probability below 2.1×10^{-5} (exact 95 % bound [10]). The booster β mismatch dominates the one-at-a-time ranking, followed by the quadrupole gradient and energy errors. A predeclared prefix-stability target of 10^{-5} on the median is not met by the 50 000-sample ensembles (changes of 1.1–2.1$\times10^{-5}$ between 25 000 and 50 000 samples) but is met by

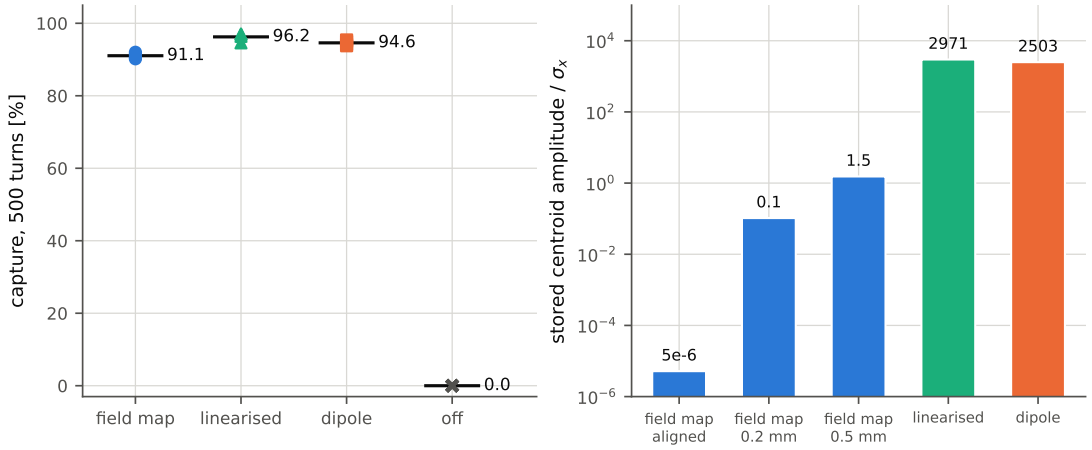


Figure 5. Paired comparison at the operating point. Left: capture of four independent 500-particle bunches (markers) and their mean (bar) for each kick model. Right: stored-beam centroid amplitude in units of the rms beam size after one passage for the NKM aligned and offset by 0.2 and 0.5 mm from the stored orbit, and for the calibrated controls (logarithmic scale).

the 400 000-sample ensemble (change of 2.4×10^{-6} between 200 000 and 400 000 samples). The pooled median is known to about 0.6 %.

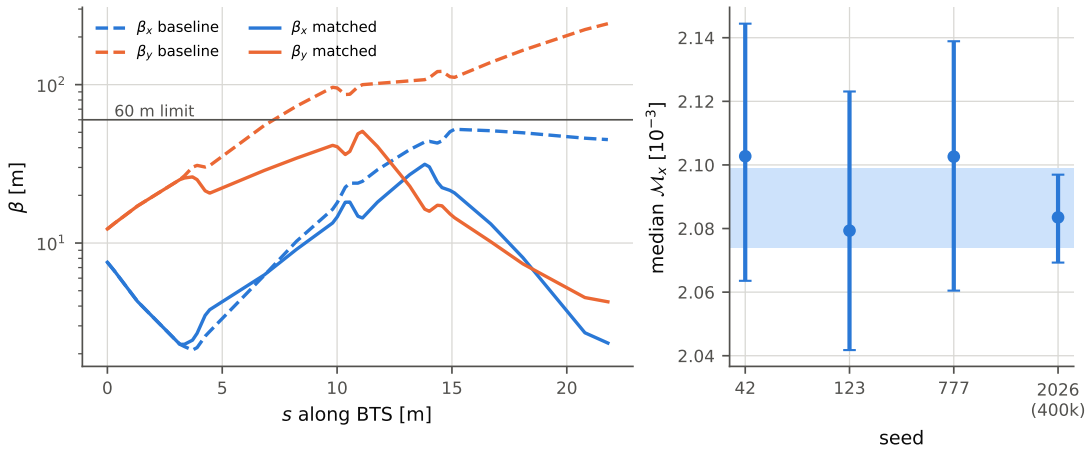


Figure 6. Left: horizontal and vertical β functions along the BTS for the baseline (dashed) and matched (solid) quadrupole settings, propagated from the booster extraction optics. Right: median horizontal exit mismatch of three independent 50 000-sample error ensembles and one 400 000-sample ensemble (seed 2026) with order-statistic 95 % intervals; the band is the pooled interval.

5.2 Injection capture under combined errors

Capture robustness is evaluated by tracking, for each error realization, a 200-particle booster bunch through the perturbed BTS (gradient, offset, roll and dipole-field errors; booster centroid jitter 0.5 mm/0.2 mrad, β mismatch 5 %, energy offset 0.1 %, emittance 10 % and energy-spread variation) into the ring with septum position (0.1 mm), NKM field scale (0.5 %), NKM alignment

(0.2 mm) and closed-orbit (0.2 mm) errors, for 500 turns. Realizations are the resampling unit. Over 150 realizations from three independent seeds the mean capture is 37.8 % (95 % cluster-bootstrap interval 33.1–42.7 %, median 34.5 %) and 3 % of the realizations exceed 90 % capture (figure 7). The realization without errors captures 88 %. The paired uniform-dipole control also degrades, from 94.6 to 59.3 % (seed 42), so most of the loss is caused by the errors; the NKM loses a further 16.5 percentage points relative to the dipole on identical beams, because its kick gradient amplifies the centroid errors. The injected-orbit error at the septum is the strongest predictor (Spearman $\rho = -0.50$ for the angle error): the BTS turns the assumed booster jitter into rms errors of 0.63 mm and 0.32 mrad at the septum, comparable to the angular window. Under the same errors the stored-beam centroid amplitude is 0.10 rms sizes (median), 1.7 (95th percentile) and at most 6.4, dominated by the kicker and closed-orbit offsets. Tripling all error amplitudes reduces capture to a few per cent, which confirms that the ensemble responds to the errors. Two diagnostic ensembles on the same 150 error realizations separate static and shot-to-shot sources. In operation the static trajectory error that 0.1 mm quadrupole offsets produce at the septum (0.7 mm rms) is removed by steering; with ideal correction of the static trajectory at the septum exit, keeping the booster centroid and energy jitter and all optics, kicker and ring errors, the mean capture is 40.9 % (36.3–45.6 %), a paired gain of only 3.0 percentage points (0.4–5.7). Removing the booster jitter instead, while leaving the static errors uncorrected, raises the mean capture to 50.8 % (45.9–55.8 %), a paired gain of 13.0 points (8.8–17.3). The shot-to-shot injected-orbit jitter is therefore the limiting error.

The thin kick misrepresents the sensitivity to this error. Repeating the 150 realizations with the thick magnet at its re-matched angle gives a mean capture of 47.4 % (42.4–52.4 %, 88.5 % without errors), a paired gain of 9.5 points (6.1–13.2) over the thin kick. The gain is confined to positive angle errors (figure 7, right). A septum angle error of +0.3 mrad raises the betatron amplitude of the kicked centroid by +1.31 mm with the thin kick but changes it by -0.12 mm with the thick magnet, whereas for −0.3 mrad both give +0.65 and +0.60 mm. A plausible reason is that a trajectory with a larger angle moves towards the axis while it is inside the field, so the kick it accumulates differs from the kick at the magnet centre that the thin model applies. Cross-runs on 100 realizations confirm that the gain is due to the magnet model rather than the injection angle: moving the thin kick to the thick-magnet angle changes capture by -1.8 points (-4.4 to +0.9), the thick magnet at the thin-kick angle gains +5.3 points (+1.7 to +9.2), and re-matching its angle adds +3.9 (+1.2 to +6.7).

6 Discussion and limitations

The results support four statements for this lattice. First, the dynamic aperture, not the kick strength, limits NKM injection, and a linear ring model cannot predict capture. Second, the useful window is defined by cancelling the incoming angle close to the kick-map extremum, where the kick gradient across the bunch is smallest; this quantifies, for this design, the efficiency mechanism identified in [5]. Third, the NKM is transparent for the stored beam, but because its field is cubic near the axis the transparency is set by the alignment of the kicker centre to the stored orbit: a 0.2 mm offset already produces a centroid oscillation of about a tenth of the rms beam size. Fourth, with an angular window of about ±0.5 mrad, the injected-orbit stability at the septum, rather than magnet errors in the transfer line, determines whether the window can be used: the assumed booster extraction jitter must be reduced, or the transfer-line optics chosen to reduce its transmission to the septum,

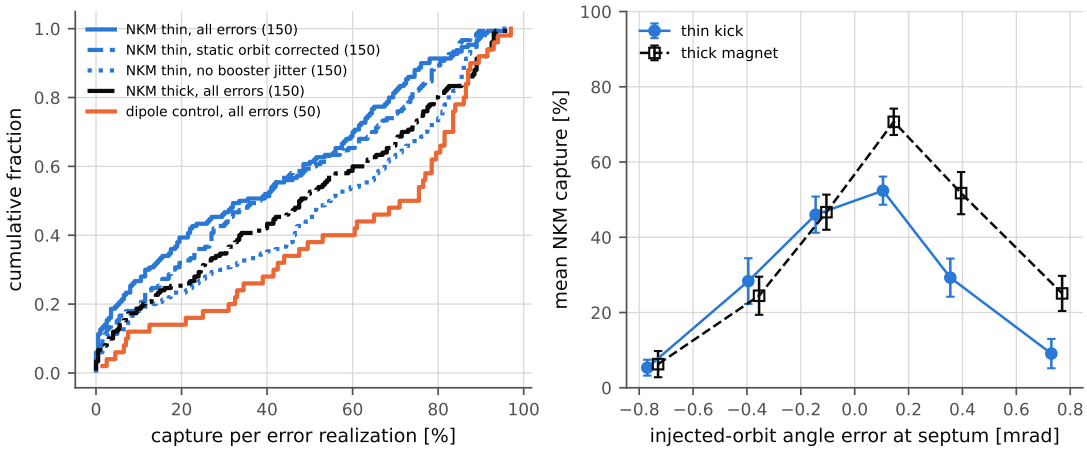


Figure 7. Combined-error robustness at the operating point. Left: cumulative distribution of capture per error realization (three seeds) for the thin-kick NKM with all errors, with the static trajectory corrected and without booster centroid jitter, for the thick-magnet NKM, and for the paired uniform-dipole control (seed 42). Right: mean capture in bins of the signed injected-orbit angle error at the septum exit for the thin kick and the thick magnet (bars: standard error).

before capture comparable to the nominal value can be expected. The thin-kick model is adequate for locating the window but not for its error sensitivity, which the thick magnet reduces for one sign of the angle error.

The limitations are explicit. The ring is tracked without magnet errors or chamber apertures; the thick-magnet model uses the midplane field profile $B_y(x)$ uniformly over the magnet length, without fringe fields or vertical dependence; the kick map is a magnetostatic calculation without measurement; the error budget is assumed rather than derived from component specifications; and capture is defined as survival over 500 turns, not as lifetime. A dedicated optimisation of the injected-beam optics at the septum and of the ring sextupoles, as done at BESSY II [3], is expected to enlarge the window and was not attempted here.

7 Conclusions

For a wire-based NKM in a 4 GeV ring with a −12 to +13 mm on-momentum dynamic aperture, nonlinear tracking with a coupled transfer line shows that angle-matched injection near the kick-map extremum captures 91.1 % of the bunch, 94.6 % for an equally strong uniform kick, with an angular tolerance of about ±0.5 mrad. The aligned kicker leaves the stored beam essentially undisturbed; transparency is limited by kicker-to-orbit alignment at the 0.1–0.2 mm level. Under the assumed combined errors the mean capture is 37.8 % with the thin kick and 47.4 % with the thick magnet, limited by the injected-orbit jitter transmitted from booster extraction; a thin-kick model underestimates the tolerance to that jitter. Linear one-turn-map models overestimate capture and should not be used to design such injection systems.

Data, software and code availability

The simulation code, configurations and analysis scripts are available in the nkm-injection repository (https://github.com/cspark7701/nkm-injection). The figures and numbers in this article are generated from the campaign artifacts listed, with SHA-256 digests, in the file `figure_provenance.csv` of the manuscript directory `docs/jinst-paper/generated`. The RADIA field map, the ring lattice and the transfer-line definition are included as input files. An archived release with a persistent identifier will be created on acceptance.

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